

# FlyByWire feature user manuals

One manual per `core/*` branch of the fork (28 branches): what each feature does, how to use it, how to test it in the sim, what to expect, and its limitations with the reason

|                         |                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aircraft</b>         | A32NX and A380X                                                                                                                                                                                                                                                     |
| <b>Where to find it</b> | This folder: each manual as HTML (with its <code>img/NN/</code> pictures) and as PDF; <code>FBW-user-manuals.pdf</code> holds them all, <code>FBW-user-manuals.docx</code> is the editable Word version                                                             |
| <b>Branch</b>           | The <code>core/*</code> branches of the fork <code>TCdr/aircraft</code> : what to open the upstream pull requests from                                                                                                                                              |
| <b>Status</b>           | Restructured on 2026-10-04 from the 31 per-PR manuals of 2026-09-28 (their content is merged into the core manuals); each feature carries its own test status.                                                                                                      |
| <b>Pictures</b>         | Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams. Features added after 2026-09-28 are described in text. |

1. The manuals

2. Stacks: what must be merged first
3. Companion changes in aircraft-large-files

4. How to read them

## 1. The manuals

| # | Manual                                                                             | Aircraft                                                             | Branch                               | Based on            | Features |
|---|------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------|---------------------|----------|
| 1 | <a href="#">Weather radar and terrain on the ND and VD (PDF)</a>                   | A32NX and A380X                                                      | <code>core/nd-weather-terrain</code> | <code>master</code> | 8        |
| 2 | <a href="#">Surveillance: SURV controls, ECAM SURV lines and TCAS (PDF)</a>        | A380X (SURV panel, VD messages, ECAM lines) and A32NX + A380X (TCAS) | <code>core/surveillance</code>       | <code>master</code> | 7        |
| 3 | <a href="#">Independent CPT and F/O MCDUs (PDF)</a>                                | A32NX                                                                | <code>core/a32nx-mcdu</code>         | <code>master</code> | 3        |
| 4 | <a href="#">ECAM procedures, EMER CANC and FIRE TEST (PDF)</a>                     | A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option)               | <code>core/ecam-fws</code>           | <code>master</code> | 5        |
| 5 | <a href="#">Runway awareness: RAAS and OANS (PDF)</a>                              | A32NX (RAAS) and A380X (OANS)                                        | <code>core/oans-btv-raas</code>      | <code>master</code> | 4        |
| 6 | <a href="#">A380X standby instruments (ISIS) (PDF)</a>                             | A380X                                                                | <code>core/isis</code>               | <code>master</code> | 4        |
| 7 | <a href="#">Lights: strobe and beacon timing, interior lighting, presets (PDF)</a> | A32NX and A380X                                                      | <code>core/lights</code>             | <code>master</code> | 2        |
| 8 | <a href="#">Fire warnings and the ENG FIRE pushbutton (PDF)</a>                    | A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve)       | <code>core/fire</code>               | <code>master</code> | 3        |

|    |                                                                          |                                                                                                                                          |                             |                                    |    |
|----|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------|----|
| 9  | <a href="#">A380X MFD FMS and SURV pages (PDF)</a>                       | A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit and the company winds is also in the A32NX FMS, unchanged for it) | core/a380x-mfd-fms          | core/surveillance                  | 16 |
| 10 | <a href="#">A380X ATC COM and datalink (PDF)</a>                         | A380X (the ACARS provider connection is shared with the A32NX)                                                                           | core/a380x-atccom-datalink  | core/a380x-mfd-fms                 | 5  |
| 11 | <a href="#">A380X fuel jettison and FINAL fuel (PDF)</a>                 | A380X                                                                                                                                    | core/fuel                   | core/a380x-mfd-fms                 | 4  |
| 12 | <a href="#">GPS receivers, GPS deselection and ECAM STATUS (PDF)</a>     | A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only                                              | core/gps                    | core/a380x-mfd-fms                 | 6  |
| 13 | <a href="#">A380X interactive ND (PDF)</a>                               | A380X                                                                                                                                    | core/a380x-interactive-nd   | core/a380x-mfd-fms                 | 1  |
| 14 | <a href="#">flyPad performance calculators (PDF)</a>                     | A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)                                 | core/efb-performance        | core/a380x-mfd-fms                 | 10 |
| 15 | <a href="#">Cockpit printers (PDF)</a>                                   | A380X (printing, buttons) and A32NX (paper door, no torn paper edge)                                                                     | core/printer                | core/a380x-atccom-datalink         | 5  |
| 16 | <a href="#">flyPad ground: taxi route, ATC frequencies and GSX (PDF)</a> | A32NX and A380X (the OANS flags are A380X only)                                                                                          | core/efb-ground             | core/efb-performance               | 5  |
| 17 | <a href="#">flyPad Material 3 redesign and independent flyPads (PDF)</a> | A32NX and A380X (the Buffet tab: A32NX only)                                                                                             | core/efb-redesign           | core/efb-ground                    | 15 |
| 18 | <a href="#">A32NX air conditioning per the FCOM (PDF)</a>                | A32NX                                                                                                                                    | core/a32nx-air-conditioning | core/gps                           | 4  |
| 19 | <a href="#">A380X cockpit buttons, lights and probes (PDF)</a>           | A380X                                                                                                                                    | core/a380x-cockpit          | master                             | 5  |
| 20 | <a href="#">A32NX exterior spoilers by panel group (PDF)</a>             | A32NX                                                                                                                                    | core/a32nx-exterior         | master                             | 1  |
| 21 | <a href="#">A380X electrical alerts and AC ESS loads (PDF)</a>           | A380X                                                                                                                                    | core/a380x-elec             | core/gps                           | 4  |
| 22 | <a href="#">A380X flyPad failures (PDF)</a>                              | A380X                                                                                                                                    | core/failures-a380x         | master + core/fire + core/ecam-fws | 14 |

|    |                                                                 |                 |                            |                                              |    |
|----|-----------------------------------------------------------------|-----------------|----------------------------|----------------------------------------------|----|
| 23 | <a href="#">A32NX flyPad failures (PDF)</a>                     | A32NX           | core/failures-a32nx        | core/gps + core/fire                         | 14 |
| 24 | <a href="#">Display unit failures and reconfiguration (PDF)</a> | A32NX and A380X | core/display-unit-failures | core/nd-weather-terrain                      | 4  |
| 25 | <a href="#">A380X weather radar GAIN, ELEVN and TILT (PDF)</a>  | A380X           | core/a380x-wxr-controls    | core/nd-weather-terrain + core/a380x-mfd-fms | 4  |
| 26 | <a href="#">A380X VD runways as flat ground (PDF)</a>           | A380X           | core/a380x-vd-runways      | core/nd-weather-terrain + core/oans-btv-raas | 1  |
| 27 | <a href="#">FCOM audit fixes (PDF)</a>                          | A32NX and A380X | core/fcom-audit-fixes      | master                                       | 7  |
| 28 | <a href="#">RAT ground stow (PDF)</a>                           | A32NX and A380X | core/rat-ground-stow       | master                                       | 1  |

## 2. Stacks: what must be merged first

Most branches are based directly on `master` (FlyByWire master 255546cb0 merged in) and can be merged in any order. The others are stacked: a branch needs every branch under it merged first.

| Stack                           | Merge order                                                                                                                                                                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A380X MFD                       | core/surveillance > core/a380x-mfd-fms > then any of core/a380x-atccom-datalink, core/fuel, core/gps, core/a380x-interactive-nd, core/efb-performance                                                                                                                                                 |
| Printer                         | ... core/a380x-atccom-datalink > core/printer                                                                                                                                                                                                                                                         |
| flyPad                          | ... core/efb-performance > core/efb-ground > core/efb-redesign                                                                                                                                                                                                                                        |
| GPS                             | ... core/gps > core/a32nx-air-conditioning, core/a380x-elec, core/failures-a32nx (which also merges core/fire)                                                                                                                                                                                        |
| A380X failures                  | core/fire + core/ecam-fws > core/failures-a380x (merged in 2026-10-05 for the engine failures)                                                                                                                                                                                                        |
| ND                              | core/nd-weather-terrain > core/display-unit-failures; core/nd-weather-terrain + core/a380x-mfd-fms > core/a380x-wxr-controls; core/nd-weather-terrain + core/oans-btv-raas > core/a380x-vd-runways                                                                                                    |
| Independent (on master)         | core/a32nx-mcdu, core/ecam-fws, core/oans-btv-raas, core/isis, core/lights, core/fire, core/a380x-cockpit, core/a32nx-exterior, core/fcom-audit-fixes, core/rat-ground-stow                                                                                                                           |
| Feature split over two branches | The RAT stow row of core/efb-redesign needs core/rat-ground-stow (the systems side): merge both for the feature to work                                                                                                                                                                               |
| Same panel.cfg section          | core/display-unit-failures (A32NX DuMessage gauge) and core/efb-redesign (A32NX F/O flyPad) both add [VCockpit21]: whichever is merged second rennumbers its section to [VCockpit22]                                                                                                                  |
| Shared engine failure module    | core/failures-a32nx and core/failures-a380x both carry the same fbw-common engine failure files (engine_failure.rs, ata.ts): whichever is merged second keeps one copy; in failures/mod.rs and the A380X a380_systems_wasm failure list keep the GPS entries of core/gps followed by one ATA 72 block |

### 3. Companion changes in aircraft-large-files

Some features change the 3D cockpit models, which live in the separate FlyByWire repository `aircraft-large-files`. Their companion commits are on the fork `TCdr/aircraft-large-files` and must be merged with the core branch.

| Companion commit | What it adds                                                                                                                        | Core branch                |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 5563e31          | A32NX MCDU screens: both MCDUs on one texture (UV halves)                                                                           | core/a32nx-mcd�            |
| 36b4bd9          | A380X pedestal printer paper                                                                                                        | core/printer               |
| 27d6123          | A380X printer door, roller unit and paper slit                                                                                      | core/printer               |
| 99237fb          | A32NX printer door                                                                                                                  | core/printer               |
| 24f7b02          | A380X F/O flyPad screen (independent flyPads)                                                                                       | core/efb-redesign          |
| d3060ea          | A32NX F/O flyPad screen (independent flyPads)                                                                                       | core/efb-redesign          |
| 8c14d21          | A380X printer ON light (green window)                                                                                               | core/printer               |
| fa6c5d1          | A380X fuel jettison plume at the outlets (stacked on 8c14d21)                                                                       | core/fuel                  |
| b8fb739          | A380X MFP (multi-function probe) vane animations                                                                                    | core/a380x-cockpit         |
| b790e8d          | A32NX DU reconfiguration: 17 overlay glasses on the EIS screens, DU_MESSAGE and DU_BLANK materials (stacked on 99237fb and d3060ea) | core/display-unit-failures |

### 4. How to read them

- Each manual has one numbered section per **feature** of its branch. The grey line under the feature title gives the aircraft, where to find it and its **status** (sim-tested, or not, with the date).
- **What it is for**: the problem the change solves, in a few sentences.
- **How to use**: numbered steps for normal use.
- **How to test**: a table of steps a tester follows in the sim, each with the expected result. If the sim does something else, it is a bug worth reporting with the branch name.
- **What you should see** tables: the expected behaviour in each situation; **Limitations and why**: what does not work yet, the reason and what to do about it.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.